time-to-go equals 12 seconds at an altitude of about 150 feet. The automatic landing phase program P65 initiates an automatic descent by nulling the horizontal velocity relative to the surface and maintaining the rate of descent at 3 ft/sec. The manual landing phase P66 is initiated when the crew changes the position of the primary guidance mode control switch from automatic to attitude-hold and then actuates the rate-of-descent control switch. Vehicle attitude changes are then controlled manually by the crew, the descent engine throttle is under computer control, and the Commander can introduce 1-ft/sec increments in the descent rate using the rate-of-descent switch.

Throughout the descent, maximum use was made onboard, as well as on the ground, of all data, system responses, and cues, based on vehicle position with respect to designated lunar features, to assure proper operation of the onboard systems. The two onboard guidance systems provided the crew with a continuous check of selected navigation parameters. Comparisons were made on the ground between data from each of the onboard systems and comparable information derived from tracking data. A powered flight processor was used to simultaneously reduce Doppler tracking data from three or more ground stations and calculate the required parameters. A filtering technique was used to compute corrections to the Doppler tracking data and thereby define an accurate vehicle state vector. The ground data were used as a voting source in case of a slow divergence between the two onboard systems.

## 5.2 PREPARATION FOR POWERED DESCENT

The crew entered and began activation of the lunar module following the first sleep period in lunar orbit (see section 4.8). A listing of significant events for lunar module descent is presented in table 5-I.

Undocking was accomplished on schedule just prior to acquisition of signal on lunar revolution 13. After the lunar module inspection by the Command Module Pilot, a separation maneuver was performed by the command and service modules, and 20 minutes later, the rendezvous radar and VHF ranging outputs were compared. The two systems agreed and indicated 0.7-mile in range. The inertial measurement unit was aligned optically for the first time, and the resulting gyro torquing angles were well within the platform drift criteria for a satisfactory primary system. Descent orbit insertion was performed on time approximately 8 minutes after loss of Network line-of-sight. Table 5-II contains the trajectory information on descent orbit insertion, as reported by the crew following acquisition of signal on revolution 14. A relatively large Z-axis residual for the abort guidance system was caused by an incorrectly loaded target vector. With this exception, the residuals were well within the three-sigma dispersion (plus or minus 0.6 ft/sec) predicted before flight.

Commander again switched from automatic to attitude-hold control and manually increased the flight-path angle by pitching to a nearly vertical attitude for range extension. Manual control began at an altitude of approximately 600 feet. Ten seconds later, at approximately 400 feet, the rate-of-descent mode was activated to control descent velocity. In this manner, the spacecraft was guided approximately 1100 feet downrange from the initial aim point.

Figure 5-3 contains histories of altitude compared with altitude-rate from the primary and abort guidance systems and from the Network powered flight processor. The altitude difference existing between the primary system and the Network at powered descent initiation can be observed in this figure. All three sources are initialized to the primary guidance state vector at powered descent initiation. The primary system, however, is updated by the landing radar, and the abort guidance system is not. As indicated in the figure, the altitude readouts from both systems gradually diverge so as to indicate a lower altitude for the primary system until the abort system was manually updated with altitude data from the primary system.

The powered flight processor data reflect both the altitude and downrange errors existing in the primary system at powered descent initiation. The radial velocity error is directly proportional to the downrange position error such that a 1000-foot downrange error will cause a 1-ft/sec radial velocity error. Therefore, the 20 000-foot downrange error existing at powered descent initiation was also reflected as a 20-ft/sec radial velocity residual. This error is apparent on the figure in the altitude region near 27 000 feet, where an error of approximately 20 ft/sec is evident. The primary-system altitude error in existence at powered descent initiation manifests itself at touchdown when the powered flight processor indicates a landing altitude below the lunar surface. Figure 5-4 contains a similar comparison of lateral velocity from the three sources. Again, the divergence noted in the final phases in the abort guidance system data was caused by a lack of radar updates.

Figure 5-5 contains a time history of vehicle pitch attitude, as recorded by the primary and abort guidance systems. The scale is set up so that a pitch of zero degrees would place the X-axis of the vehicle vertical at the landing site. Two separate designations of the landing site are evident in the phase after manual takeover. Figure 5-6 contains comparisons for the pitch and roll attitude and indicates the lateral corrections made in the final phase.

Figure 5-7 is an area photograph, taken from a Lunar Orbiter flight, showing the landing site ellipse and the ground track flown to the landing point. Figure 5-8 is an enlarged photograph of the area adjacent to the lunar landing site and shows the final portions of the ground track to landing. Figure 5-9 contains a preliminary attempt at reconstructing the surface terrain viewed during descent, based upon trajectory and radar